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A MINI PROJECT REPORT

On

Broject — AI-powered developer collaboration matcher

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Advanced Artificial Intelligence
In
Department of CSE-AIML (SEM VIII)

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Abstract

This project presents an AI-driven Developer Collaboration and Project Recommendation System designed to intelligently match software developers based on their skills, experience, and project preferences by leveraging advanced natural language processing and machine learning techniques. The system utilizes sentence embeddings generated through transformer models along with clustering algorithms such as Gaussian Mixture Models (GMM) to group developers into meaningful skill clusters, enabling efficient identification of compatible collaborators. User profiles are constructed using attributes including programming languages, technical domains, experience levels, work styles, and availability, and a hybrid scoring mechanism combining cosine similarity and contextual factors is applied to rank potential matches. Furthermore, the system integrates generative AI to propose innovative project ideas tailored to the strengths of matched developers, enhancing creativity and collaboration outcomes. A Streamlit-based interface provides an interactive environment where users can input their profiles, receive cluster assignments, explore recommended collaborators, and generate project concepts in real time. By improving skill-based networking, optimizing team formation, and fostering innovation in software development ecosystems, this solution directly contributes to Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by promoting technological advancement and building intelligent digital collaboration infrastructure, while also supporting Sustainable Development Goal 8 (Decent Work and Economic Growth) through enhanced productivity and opportunities in the technology sector.

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Chapter 1
Introduction

1.1 Fundamentals

This section introduces the fundamental concepts of the proposed AI-based Developer Collaboration and Recommendation System. The project is based on machine learning and natural language processing techniques that enable intelligent matching of developers according to their skills, experience, and preferences. As seen in the implementation, sentence transformer models are used to generate embeddings from developer skill sets, which convert textual data into numerical vectors for analysis. These embeddings are further processed using clustering algorithms such as Gaussian Mixture Models (GMM) to group developers into similar skill clusters. Additionally, cosine similarity is applied to measure compatibility between users within the same cluster. The system also incorporates generative AI models to suggest project ideas based on combined skill sets. Overall, the fundamentals of this project lie in combining NLP, clustering, similarity measures, and recommendation techniques to build an intelligent and scalable collaboration platform.

1.2 Objectives

The objective of this work is to design and develop an intelligent developer matching and project recommendation system that improves collaboration efficiency. The system aims to analyze developer profiles, including skills, domains, and experience levels, and group them using clustering techniques to identify similar users. It focuses on implementing feature extraction using sentence embeddings to accurately represent user skills and applying similarity measures to recommend compatible collaborators. Finally, the system aims to provide an interactive platform (as shown in the Streamlit interface where users can easily input their data, receive recommendations, and explore collaboration opportunities effectively.

Chapter 2
Literature Survey

2.1 Introduction

This chapter presents a review of existing research and methodologies related to recommendation systems, machine learning, and developer collaboration platforms. Recommendation systems have evolved significantly with the integration of advanced techniques such as natural language processing (NLP), deep learning, and hybrid models that combine multiple approaches. The purpose of this chapter is to analyze various state-of-the-art methods used in recommendation systems, understand their strengths and limitations, and identify research gaps that motivate the development of the proposed system. The reviewed works focus on techniques such as collaborative filtering, content-based filtering, hybrid recommendation models, and the use of embeddings and attention mechanisms.

2.2 Literature Review

Several research works have contributed to the advancement of recommendation systems using different machine learning and deep learning techniques.

1.Xiang Wang, Xiangnan He, Meng Wang, and Fuli Feng (2024) proposed a recommendation system that integrates sentence transformers with graph-based approaches to enhance recommendation accuracy. Their model utilizes semantic embeddings along with user-item interaction graphs to improve performance on large-scale datasets such as Amazon and Yelp. The results demonstrate significant improvement in recommendation accuracy; however, the approach requires large-scale graph data and involves high computational complexity, making it less suitable for lightweight applications.

2.Yongfeng Zhang, Xu Chen, Qingyao Ai, and Yiqun Liu (2024) introduced a hybrid recommendation system that combines large language models (LLMs) with collaborative filtering techniques. This approach improves personalization and effectively addresses the cold-start problem by leveraging contextual reasoning capabilities of LLMs.

2.3 Literature Summary

Author & Year	Techniques Used	Dataset	Key Result	Limitations
Xiang Wang, Xiangnan He, Meng Wang (2024) [1]	Sentence Transformers + interaction-aware embeddings	Multi-domain recommendation datasets	Outperforms traditional collaborative filtering	Requires large training data
Yongfeng Zhang, Xu Chen, Qingyao Ai, Yiqun Liu (2024) [2]	LLM + Collaborative Filtering	Real-world recommendation datasets	Improves performance by combining LLM	Higher computational cost; depends on pretrained CF
Chenyang Wang, Hongyu Zhao, Yong Liu (2023) [3]	Sequential recommendation + attention mechanisms	Scientific paper datasets	Captures dynamic user intent	Limited generalization beyond academic
Jianxun Lian, Xiaohuan Zhou (2025) [4]	Multi-interest learning + LLM-based	Benchmark + industrial datasets	Captures diverse user interests	Complexity in modeling multiple interests.
Zheng Liu, Yukun Chen, Zhaochun Ren (2023) [5]	content-based + collaborative	Review-based datasets	Better recommendation quality using sentiment analysis	Domain-dependent and requires extensive preprocessing

Table 2.3 Literature survey summary

Chapter 3

Implementation Details

3.1 Overview

This chapter presents the overall design and implementation of the proposed AI-based Developer Collaboration and Recommendation System. The system is designed to intelligently match developers based on their skills, experience, and preferences, and to generate suitable project ideas. It integrates machine learning, natural language processing (NLP), clustering techniques, and generative AI to provide an efficient and scalable solution. The chapter describes both the existing methodologies and the proposed approach, followed by detailed implementation steps.

3.1.1 Proposed Methodology and System

The proposed system overcomes these limitations by integrating advanced AI techniques. It uses sentence transformer models to convert developer skill sets into vector embeddings, enabling semantic understanding of skills. These embeddings are then clustered using Gaussian Mixture Models (GMM) to group developers into similar categories.

An interactive user interface is developed using Streamlit, allowing users to input their profiles, view recommended collaborators, and generate project ideas in real time. This approach ensures scalability, better accuracy, and enhanced user experience compared to traditional systems.

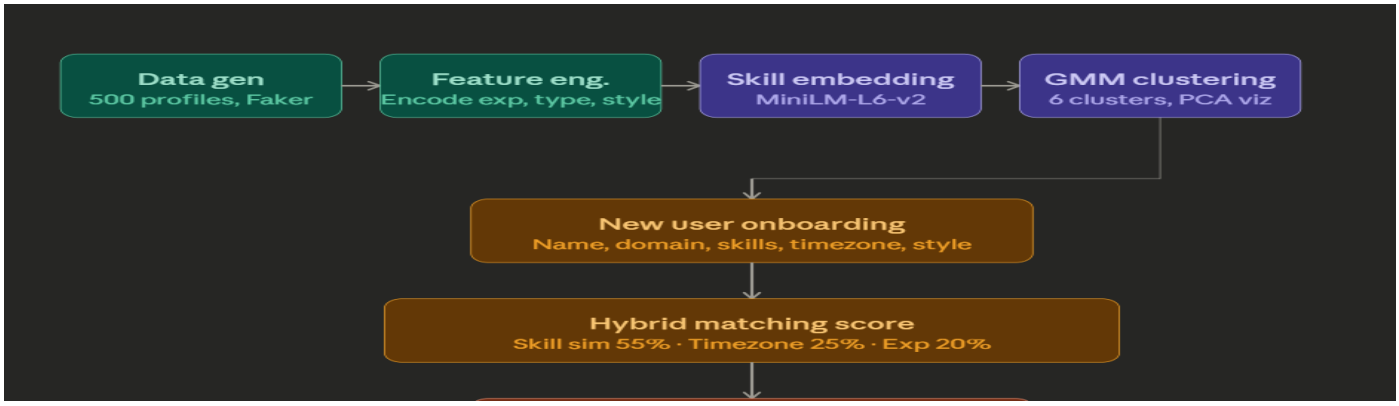

Fig_1: Proposed System

3.2 Implementation Details

The system is implemented using Python and various machine learning libraries. The workflow includes data preprocessing, embedding generation, clustering, recommendation, and user interface integration. The

implementation is modular, ensuring easy maintenance and scalability.
Fig_2 shows the user interface of the AI Collaboration Matcher developed using Streamlit. In this interface, users enter their profile details such as name, primary domain, technical skills, and experience level. Fig_3 displays the output of the system after processing the user input. It presents the top 5 matched developers along with their compatibility scores.

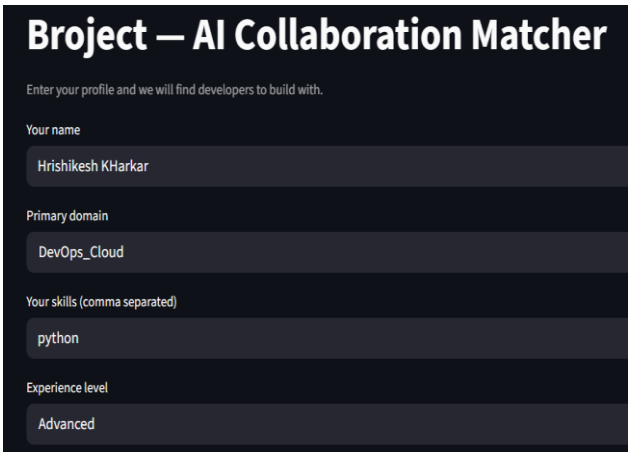

Fig_2: Broject Dashboard

Fig_3: Matchmaking Clustering

Chapter 4

Project Inputs and Outputs

4.1 Inputs Details

The system accepts user input through an interactive interface where developers provide their profile information. As shown in Fig_2 the key inputs include the user's name, primary domain (such as DevOps, Web Development, AI/ML), technical skills entered as comma-separated values, and experience level (Beginner, Intermediate, or Advanced).

4.2 Output Details and Screenshots

The output of the system is displayed in the Fig_4 organized and user-friendly format. After entering the input details and clicking the "Find My Matches" button, the system generates the top matches along with their compatibility scores.

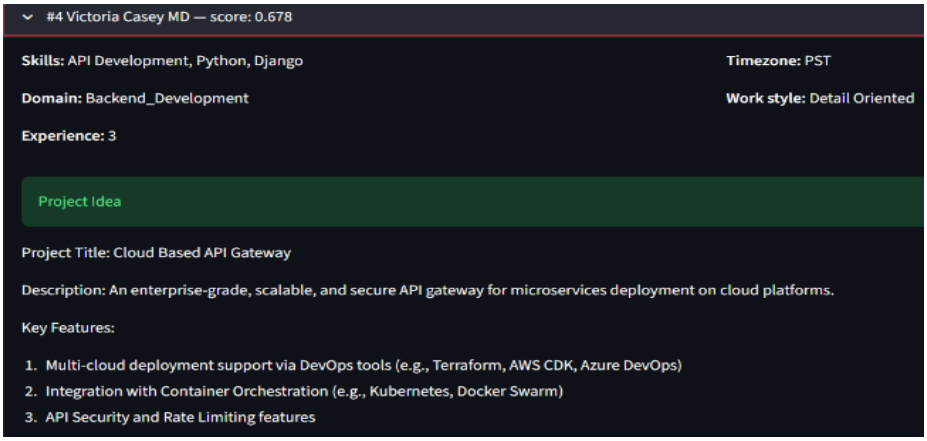

Fig_4: Generative Idea generation

Summary and Future Scope

1.1 Summary

This project presents an AI-based Developer Collaboration and Recommendation System designed to intelligently match developers based on their skills, experience, and preferences. The system utilizes natural language processing techniques to convert textual skill data into embeddings, enabling semantic understanding of user profiles. Clustering algorithms such as Gaussian Mixture Models are applied to group developers into similar categories, while cosine similarity and hybrid scoring methods are used to identify and rank compatible collaborators. The system also integrates generative AI to suggest innovative project ideas based on the combined skills of matched users. An interactive Streamlit interface enhances usability by allowing real-time input and recommendation display. Overall, the project successfully demonstrates an efficient, scalable, and intelligent approach to improving developer collaboration and project discovery.

1.2 Future Scope

The proposed system can be further enhanced in several ways to improve its performance and applicability. Firstly, the system can be extended to use real-world large-scale datasets from platforms such as GitHub or LinkedIn to improve accuracy and reliability. Advanced deep learning models and graph-based recommendation techniques can be incorporated to better capture complex relationships between developers. The inclusion of real-time collaboration features, such as chat or team formation tools, can further enhance usability. Additionally, the system can be improved by integrating feedback mechanisms where users can rate recommendations, allowing continuous learning and optimization.

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